

## Gregory DiCristofaro

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### TECHNOLOGY EXPERIENCE:

**Software Engineer:** Sleuth Kit Labs, Somerville, MA

*February 2020 - Present*

- Maintain and develop features for Digital Forensics and Incident Response software: Cyber Triage, Autopsy, and The Sleuth Kit
- Develop and maintain Java desktop applications with Swing, JavaFX, Maven, Apache Ivy, Apache POI, SQLite, PostgreSQL, and Netbeans RCP
- Manage automated testing using Jenkins, Linux, Docker, Bash, Python, Apache Ant, JUnit, and Mockito
- Installation management using Advanced Installer, shell installation scripts, Snap packages, and Homebrew
- Use Python to automate database creation, web scrape for data aggregation, verify E2E regression tests fuzzy database diffing
- Manage data pipeline processing using Java, C++, Python, cloud updatable rules over GCP
- Create RESTful and Reactive web applications using Spring Boot, Jackson, Spring JDBC, Maven, React, Material UI, Angular, Typescript, Server Sent Events, Project Reactor, Styled Components, and Swagger
- Create server side and cloud infrastructure using Docker, JMS, ActiveMQ, AWS DynamoDb, AWS Lambda, GCP Cloud Functions, GCP BigQuery, GCP Datastore
- Presenter for OSDfCon 2021 (Autopsy Scoring: Finding the Relevant Data with Analysis Results)

**Software Engineer:** KBR, Lexington Park, MD

*July 2017 - February 2020*

- Develop RESTful applications utilizing .NET Web API and Json.NET
- Create RESTful client applications in Java utilizing GSON and in C# utilizing Json.NET
- Web-based UI development with ASP.NET Web Forms, ASP.NET MVC, Kendo UI, Bootstrap, JQuery, CSS, and HTML
- Data visualizations with JavaScript, TypeScript, D3.js, and SVG
- Database interaction with Microsoft SQL Server, T-SQL, Entity Framework, Linq to SQL, and ADO.NET
- Develop web applications utilizing React, Redux, and Bootstrap frontend with ASP.NET Core backend
- Modernize J2EE web applications built with JSP and JSF

**M.S. Software Development:** Boston University, Boston, MA

*Fall 2017 - Spring 2019*

Classes in Python Development, Agile SDLC, Data Structures and Algorithms, Database Design and Implementation, Server-Side Development, and Angular Development. GPA 3.96/4.00

**Software Engineering Certificate:** Harvard University, Cambridge, MA    *Fall 2012 - Fall 2014*

Classes included Programming Microsoft .NET, Systems Programming and Machine Organization, and Software Design. GPA 4.00/4.00

**Notable Programs Built:**

**Algorithmic:**

- Scala implementation of the Seam-Carving Algorithm for content-aware image resizing
- Class scheduling program based on the AC-3 algorithm utilizing Apache POI API to import from excel documents written in Scala
- Spell-checking algorithm built in C checking words against a dictionary file utilizing a self-developed hash map with buckets
- Implementation of the Constant-Q transform in TypeScript and C++ for performing spectral analysis on music utilizing Angular development, Chart.js, Emscripten, WebAssembly, Web Audio API, and Web Workers

**Web:**

- Calendar making application developed with Node.js, Webpack, TypeScript, React, Moment.js, Material-UI, and PptxGenJS
- Python web application for converting Photoshop files into SVG's for data visualization
- Sci-fi customizable opening credit text using React, TypeScript, Material UI, Base64 encoding, and CSS animations

**TEACHING EXPERIENCE:**

**M.M. Music Education:** University of the Arts, Philadelphia, PA    *Summer 2012 - Fall 2015*

- Thesis: Designed cross-platform music composition software to foster creativity for young students that processes audio, visualizes music in a sequencer and renders standard music notation
- GPA 4.00/4.00

**Music Teacher:** St. Mary's County Public Schools, Leonardtown, MD    *August 2006 - June 2017*

- **Scheduling & Logistics:**
  - Organized grade level and school wide events
  - Determined school-wide schedule to optimize instruction and transitions
- **Leadership and Collaboration:**
  - On-site coordinator for Elementary All County Honor Band in charge of preparing public information, room reservations, personnel management, and coordinating with the director
  - Coordinated with teachers to determine scheduling for pull-out lessons in band to minimize impact on classroom teachers and coordinated with teachers to organize performances and events
  - Presenter for staff development sessions for the Fine Arts Department of St. Mary's County Public Schools